

Nasdaq-100 Top 2 Monthly Skip-Momentum Strategy

Structured strategy package for replication in other AI/backtesting tools.

This version trades the top 2 stocks, not top 1.

Includes rules, timing, return calculation, validation summary, caveats, and replication checklist.

1. Strategy Summary

Universe: Nasdaq-100 stocks.

Position count: top 2 stocks.

Portfolio weight: 50% in each selected stock.

Rebalance: monthly.

Signal timing: after month-end close.

Execution: next trading day open.

Return method: open-to-open for completed months.

Ranking score: $\text{Close}[t - 21 \text{ trading days}] / \text{Close}[t - 126 \text{ trading days}] - 1$.

2. Ranking Rule

At each month-end close, calculate the momentum score for every stock in the Nasdaq-100 universe.

Momentum Score = $\text{Close}[t - 21 \text{ trading days}] / \text{Close}[t - 126 \text{ trading days}] - 1$.

The 126-day lookback is roughly six trading months.

The 21-day skip window excludes the most recent trading month.

Rank all stocks from highest score to lowest and select the top two stocks.

3. Monthly Trading Rule

At each rebalance, hold the top 2 ranked stocks at 50% weight each.

If a selected stock remains in the top 2, keep holding it.

If a selected stock drops out of the top 2, sell it at the next monthly trade open.

Buy any new top-2 stock at the same open.

Record monthly decisions even when one or both holdings do not change.

4. Execution Timing

The strategy avoids same-day lookahead.

Month-end close signal -> next trading day open trade.

The ranking signal uses only data known as of the month-end close.

Positions are entered and exited at the next available open.

5. Return Calculation

For completed months, returns are open-to-open.

Each selected stock contributes according to its target weight.

Monthly Portfolio Return = $0.5 * \text{Stock1 Open-to-Open Return} + 0.5 * \text{Stock2 Open-to-Open Return}$.

Total Return = $\text{PRODUCT}(1 + \text{Monthly Portfolio Return}) - 1$.

6. Pseudocode

for each month:

 signal_date = last trading day before month start

 trade_date = first trading day of month

 for ticker in Nasdaq-100 universe:

 score[ticker] = close[ticker][signal_date - 21 trading days] /
close[ticker][signal_date - 126 trading days] - 1

 selected = top 2 tickers by score

 target weights = 50% each

 sell holdings not in selected at trade_date open

 buy new selected stocks at trade_date open

 hold unchanged selected stocks

monthly_return = weighted open-to-open return of selected holdings

7. Latest Annual Reset Validation

Validation universe: current Nasdaq-100 constituents.

Benchmark: QQQ.

Test period: 2020 through 2026 YTD.

Important caveat: this uses current Nasdaq-100 membership for past years, so point-in-time membership validation is still needed.

2020: Strategy 148.61%, QQQ close 45.97%, excess +102.64%, max DD -39.54%, trades 26.

2021: Strategy 38.94%, QQQ close 29.24%, excess +9.70%, max DD -50.12%, trades 18.

2022: Strategy -14.29%, QQQ close -33.22%, excess +18.93%, max DD -39.88%, trades 28.

2023: Strategy 90.50%, QQQ close 55.91%, excess +34.60%, max DD -25.98%, trades 22.

2024: Strategy 44.89%, QQQ close 27.74%, excess +17.15%, max DD -38.09%, trades 20.

2025: Strategy 145.92%, QQQ close 21.01%, excess +124.91%, max DD -52.49%, trades 18.

2026 YTD: Strategy 235.72%, QQQ close 15.77%, excess +219.95%, max DD -25.62%, trades 6.

8. Compounded Reset Return

2020-2026 YTD compounded reset return:

Strategy: 6647.15%.

QQQ adjusted-close benchmark: 251.51%.

QQQ open-to-open benchmark: 241.34%.

9. Generated Validation File

Validation report: C:\Users\skydiver1118\Documents\New
project\reports\nasdaq100_top2_skip21_monthly_o2o_validation_2020_2026ytd.md

Validation CSV: C:\Users\skydiver1118\Documents\New
project\reports\nasdaq100_top2_skip21_monthly_o2o_validation_2020_2026ytd.csv

10. Critical Caveats

Survivorship/index-membership bias: the test uses the current Nasdaq-100 constituent list for past dates.

A cleaner historical backtest should use point-in-time Nasdaq-100 membership.

Yahoo Finance adjusted open/close data can be noisy around corporate actions.

No slippage, bid/ask spread, commission, market impact, or tax impact is included.

The strategy holds only two stocks, so concentration risk is high.

11. Replication Checklist

Use daily adjusted open and adjusted close prices.

Prefer point-in-time Nasdaq-100 membership.

Compute the score only after month-end close.

Trade only at the next trading day open.

Select the top 2 ranked stocks.

Weight both selected stocks at 50%.

Calculate completed-month returns open-to-open.

Compound monthly returns to reconcile total return.

Compare against QQQ adjusted-close and/or QQQ open-to-open benchmark returns.